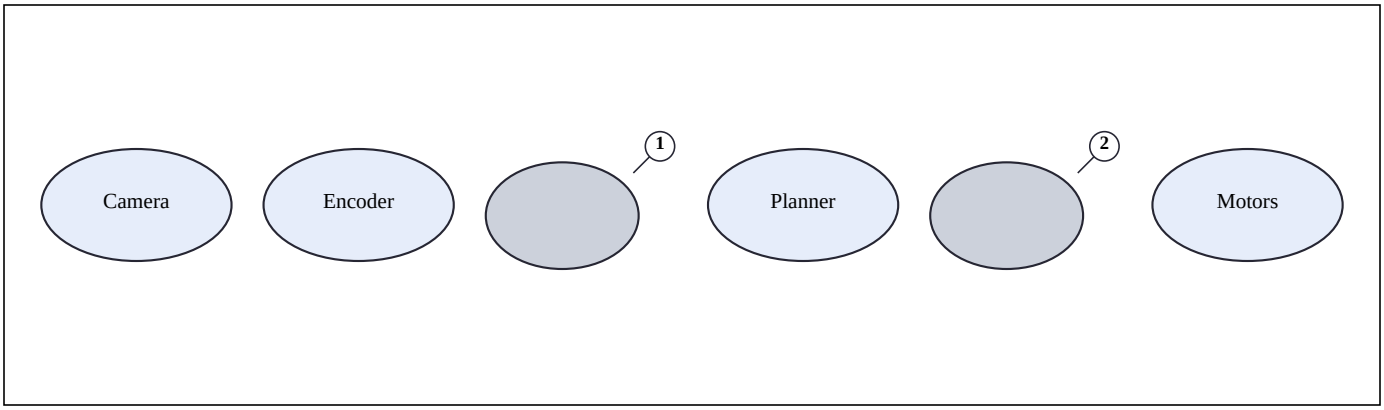

Figure 1: Overview of the delivery robot software. Images flow from the camera through the encoder and the planner to the motors.

1 Introduction

Small delivery robots now share pavements with pedestrians in several cities. They must recognise kerbs, crossings and people from a single forward camera. Most of the software runs on a low power board inside the robot body. This paper describes the perception stack of one such robot and the tests we ran on it.

The stack has four stages that run once per camera frame. A compact encoder turns each frame into a grid of features. The planner then chooses a short path and sends wheel speeds to the motors. Two helper processes watch the stack and restart it when a stage stops responding.

2 Field tests

We drove the robot for three weeks on a university campus with mixed foot traffic. Every run was recorded so that failures could be replayed and analysed later. The robot completed most deliveries without help from a remote operator. Interventions happened mainly at crossings with faded paint and at busy entrances.

Table 1: Field test summary.

Week	Runs	Km	Help
1	142	61	19
2	171	78	11
3	188	85	7

Interventions fell from week to week as we tuned the crossing detector. The largest gain came from adding images taken at dusk to the training data. Rain remained difficult because drops on the lens blur the lower part of the image.

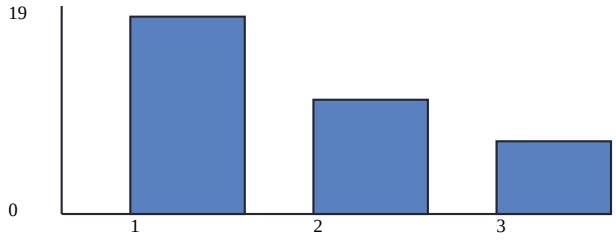

Figure 2: Remote interventions per week. Interventions fell by more than half over the three weeks.

The helper processes restarted a stalled stage nine times during the trial. Each restart took less than two seconds and the robot stopped safely while it waited. We now log the memory use of every stage to find the cause of these stalls.